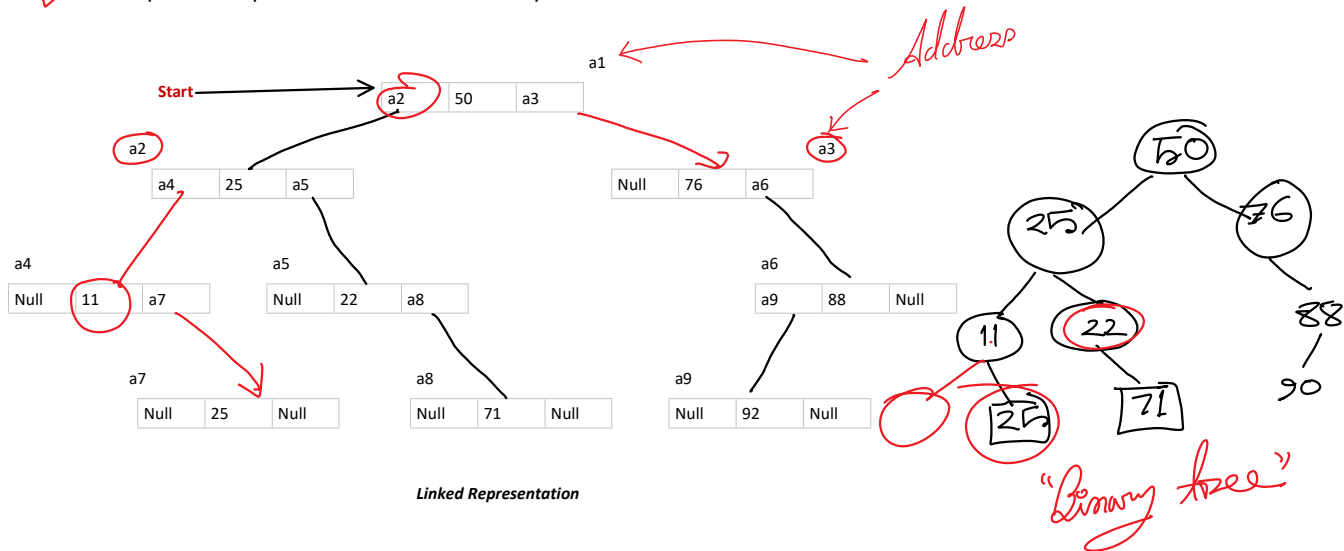


Representing Binary Tree in Computer Memory

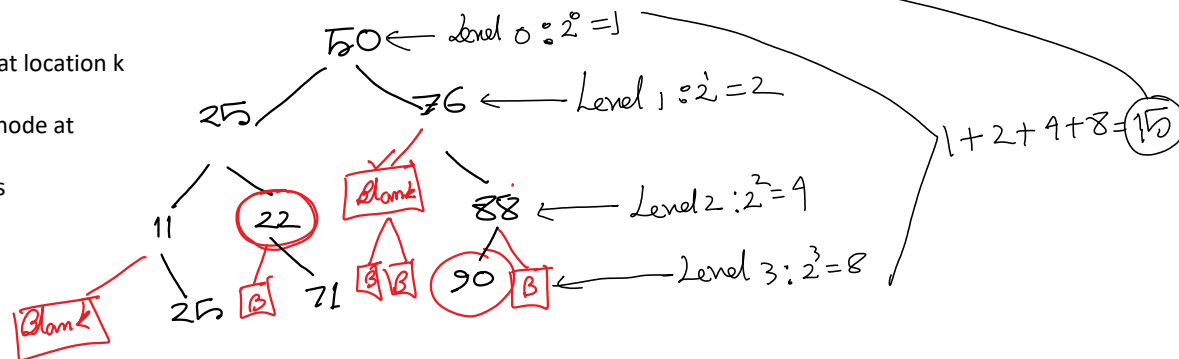
There are two ways we can represent the binary tree in the computer memory. They are:

- ✓ 1. Linked Representation in memory
- ✓ 2. Sequential representation in the memory



1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
50	25	76	11	22		88		25		71			90	

1. Left and right position of a node at location k are $2k$ and $(2k)+1$
2. Root node location number of a node at location $k = \lfloor k/2 \rfloor$
3. Here the third bracket represents the floor.



Root Node = 3
Left child = $3 \times 2 = 6$, Right child = $(3 \times 2) + 1 = 7$

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
50	25	76	11	22	✓	88	✓	25	✓	71			90	

